CS 506: Introduction to Quantum Computing

Scribe Notes: Simon's Problem and the Quantum Fourier Transform

Scribe: Shashank Pidugu

1 Simon's Problem: Inputs Come in Pairs

In Simon's problem we are given a function $f : \{0,1\}^n \to \{0,1\}^n$ with a hidden string s such that f takes the same value on x and $x \oplus s$. This means the inputs can be partitioned into groups of 2, and each group has the form

$$\{x, x \oplus s\}.$$

We call the first element the **leader** and the second the **follower**. Let I be the set of all such groups:

$$I = \{ \{x, x \oplus s\} \}.$$

There are $½ \times 2^n$ groups, so

$$|I| = 2^{n-1}.$$

Each group contributes one leader and one follower, and f gives the same output on both.

This pairing is the structure the algorithm exploits.

2 The Orthogonal Set S⊥

Define

$$S\perp = \{ z \in \{0,1\}^n : z \cdot s = 0 \pmod 2 \}.$$

S⊥ has 2^{n-1} elements. The goal of the algorithm is to sample a binary string from S⊥ **randomly, uniformly, and independently** each time it runs.

The key identity (applied to the normalized pair state) is

$$H^{\otimes n} ((1/\sqrt{2})(|x\rangle + |x \oplus s\rangle)) = (1/\sqrt{(2^{n-1})}) \Sigma_{z \in S\perp} (-1)^{x \cdot z \,(\mathrm{mod}\ 2)} |z\rangle.$$

Only strings z in S⊥ survive, and each carries a sign $(-1)^{x \cdot z}$.

Why only S⊥ survives: $H^{\otimes n}|x\rangle$ puts amplitude $(-1)^{x \cdot z}$ on every $|z\rangle$, and the same holds for $x \oplus s$. For a given z the two amplitudes add to $(-1)^{x \cdot z}(1 + (-1)^{s \cdot z})$, which is $2(-1)^{x \cdot z}$ when $s \cdot z = 0$ and exactly 0 otherwise. Every z outside S⊥ cancels (destructive interference).

3 The Oracle U_f

The function f is accessed through the unitary

$$U_f |x\rangle|b\rangle = |x\rangle|b \oplus f(x)\rangle.$$

The first register holds the input and is left unchanged. The second register is XOR'd with f(x).

This is the same reversible-gate trick as in Class 2: keep the input and XOR the output into an extra register, so no information is lost.

4 The Circuit

The circuit uses two registers of n qubits each. The top register (holding x) and the bottom register (holding b) both start in $|0\rangle$. The state at each time step is:

Time	State	What happens just before
t_0	$\|\psi_0\rangle$	Initial state: both registers in $\|0\rangle$
t_1	$\|\psi_1\rangle$	Hadamard applied to every qubit of the top register
t_2	$\|\psi_2\rangle$	U_f applied to both registers
t_3	$\|\psi_3\rangle$	Bottom register measured in the standard basis (**optional**)
t_4	$\|\psi_4\rangle$	Hadamard applied to every qubit of the top register again

At the end, the top register is **measured in the standard basis**, and this measurement is the output.

5 Following the State Through U_f

Start with

$$|\psi_0\rangle = |0^n\rangle|0^n\rangle.$$

The Hadamards create a uniform superposition on the top register:

$$|\psi_1\rangle = (H^{\otimes n}|0^n\rangle)|0^n\rangle = \left((1/\sqrt{(2^n)}) \, \Sigma_{z \in \{0,1\}^n} |z\rangle \right) |0^n\rangle.$$

Applying U_f:

$$\begin{aligned} |\psi_2\rangle &= U_f|\psi_1\rangle \\ &= (1/\sqrt{(2^n)}) \, \Sigma_z \, U_f(|z\rangle|0^n\rangle) \\ &= (1/\sqrt{(2^n)}) \, \Sigma_z |z\rangle \, |0^n \oplus f(z)\rangle \\ &= (1/\sqrt{(2^n)}) \, \Sigma_z |z\rangle \, |f(z)\rangle, \end{aligned}$$

since $0^n \oplus f(z) = f(z)$. Now regroup the sum using the pairs from Section 1. Each pair $\{x, x \oplus s\}$ shares the same value f(x), so

$$|\psi_2\rangle = (1/\sqrt{(2^{n-1})})\ \Sigma_{x \in I}\ (\ (1/\sqrt{2})|x\rangle + (1/\sqrt{2})|x \oplus s\rangle\ )\ |f(x)\rangle.$$

The circled piece on the board is the $|f(x)\rangle$ register. Regrouping into pairs is what makes the next step work.

6 Measuring the Bottom Register

Measuring the second register gives a specific value $f(x_0)$ for some $x_0 \in I$. This collapses the first register onto the matching pair:

$$|\psi_3\rangle = (\ (1/\sqrt{2})|x_0\rangle + (1/\sqrt{2})|x_0 \oplus s\rangle\ )\ |f(x_0)\rangle.$$

Applying $H^{\otimes n}$ to the top register:

$$|\psi_4\rangle = H^{\otimes n}(\ (1/\sqrt{2})|x_0\rangle + (1/\sqrt{2})|x_0 \oplus s\rangle\ )\ |f(x_0)\rangle$$
$$= (\ (1/\sqrt{(2^{n-1})})\ \Sigma_{z \in S\perp}\ (-1)^{x_0 \cdot z \pmod 2}\ |z\rangle\ )\ |f(x_0)\rangle.$$

Measuring the top register gives some $z_0 \in S\perp$ with probability

$$(\ (1/\sqrt{(2^{n-1})})\ (-1)^{x_0 \cdot z_0 \pmod 2}\ )^2 = 1/2^{n-1},$$

because the sign squares to 1. Since $S\perp$ has exactly 2^{n-1} elements, every element is equally likely: the output is a **uniformly random element of S⊥**.

Why the bottom measurement is "optional": it only decides which pair the top register collapses onto. The distribution of the final output on $S\perp$ is the same for every pair, so skipping that measurement gives the same output distribution.

7 The Quantum Fourier Transform

The **Quantum Fourier Transform** of order 2^n (QFT_{2^n}) acts on a basis state as

$$QFT_{2^n}\ |x\rangle = (1/\sqrt{(2^n)})\ \Sigma_{y \in \{0,1\}^n}\ e^{2\pi i \cdot xy/2^n}\ |y\rangle.$$

Here x is the decimal value of the bit-vector $|x\rangle$ and y is the decimal value of $|y\rangle$. The input x is a binary integer of n bits,

$$x = x_{n-1}\ x_{n-2}\ \ldots\ x_1\ x_0,$$

where each x_i is 0 or 1.

The circled factor $e^{2\pi i \cdot xy/2^n}$ is a phase. Every output basis state $|y\rangle$ has the same magnitude $1/\sqrt{(2^n)}$, so x is encoded entirely in the phases.

8 Binary Fractions and Phases

To understand these phases we write numbers between 0 and 1 in binary. For $0 \leq x < 1$:

$$x = (0.x_1x_2x_3 \ldots x_k \ldots x_n \ldots)_2,$$

where the digits carry place values 2^{-1}, 2^{-2}, 2^{-3}, …, 2^{-k}, …, 2^{-n}. Multiplying by a power of 2 moves the binary point to the right:

$$2^1x = (x_1.x_2x_3x_4 \ldots)_2$$

$$2^2x = (x_1x_2.x_3x_4 \ldots)_2$$

$$2^kx = (x_1x_2 \ldots x_k.x_{k+1} \ldots)_2$$

Now look at the phase $e^{2\pi i \cdot x}$ itself:

$$e^{2\pi i \cdot x} = e^{2\pi i \cdot (0.x_1x_2 \ldots)_2}.$$

For the shifted version, split the exponent into the integer part and the fractional part:

$$e^{2\pi i \cdot (2^k x)} = e^{2\pi i \cdot (x_1 \ldots x_k.x_{k+1} \ldots)_2}$$

$$= e^{2\pi i \cdot (x_1 \ldots x_k)_2} \cdot e^{2\pi i \cdot (0.x_{k+1} \ldots)_2}$$

$$= e^{2\pi i \cdot (0.x_{k+1} \ldots)_2}.$$

The first factor disappears because for any integer α,

$$e^{2\pi i \cdot \alpha} = \cos(2\pi\alpha) + i \sin(2\pi\alpha) = 1 + 0 = 1.$$

Multiplying by 2^k slides the binary point k places to the right. The digits that cross over become an integer and contribute a factor of 1, so only the digits after the point still matter. Useful facts: $e^{2\pi i} = 1$ and $e^{\pi i} = -1$.

9 Recovering x from a Phase State

Let $\omega = 0.x_1x_2x_3 \ldots x_n$. Suppose we are given the state

$$|\psi\rangle = (1/\sqrt{(2^n)}) \, \Sigma_{y=0}{}^{2^n-1} \, e^{2\pi i \cdot \omega y} \, |y\rangle$$

as a black box. The task is to **figure out** $x_1, x_2, \ldots, x_n$.

The case n = 1. Here $\omega = (0.x_1)_2 = x_1/2$, and

$$|\psi\rangle = (1/\sqrt{2}) \, \Sigma_{y=0}{}^{1} \, e^{2\pi i \cdot (0.x_1)_2 \cdot y} \, |y\rangle$$

$$= (1/\sqrt{2}) \, \Sigma_{y=0}{}^{1} \, e^{2\pi i \cdot (x_1/2) \cdot y} \, |y\rangle$$

$$= (1/\sqrt{2}) \, \Sigma_{y=0}{}^{1} \, e^{\pi i \cdot x_1 y} \, |y\rangle$$

$$= (1/\sqrt{2}) \, \Sigma_{y=0}{}^{1} \, (-1)^{x_1 y} \, |y\rangle$$

$$= (1/\sqrt{2})(\, |0\rangle + (-1)^{x_1} \, |1\rangle \,) = H|x_1\rangle.$$

Applying H again recovers the bit:

$$H|\psi\rangle = H(H|x_1\rangle) = |x_1\rangle.$$

For one qubit, the QFT is just the Hadamard gate, and H is its own inverse. Undoing the transform reveals x_1. The general case is handled by the product form in the next section.

10 Product Form of the Phase State

Claim. Let $\omega = (0.x_1x_2 \ldots x_n)_2 = \Sigma_{i=1}^{n} 2^{-i}x_i$. Then

$$|\psi\rangle = (1/\sqrt{(2^n)})\ \Sigma_{y=0}^{2^n-1}\ e^{2\pi i\cdot\omega y}|y\rangle$$
$$= (\ (|0\rangle + e^{2\pi i\cdot(2^{n-1}\omega)}|1\rangle)/\sqrt{2}\ )\ \otimes\ \ [\text{1st qubit}]$$
$$(\ (|0\rangle + e^{2\pi i\cdot(2^{n-2}\omega)}|1\rangle)/\sqrt{2}\ )\ \otimes\ \ [\text{2nd qubit}]$$
$$\ldots\ \otimes$$
$$(\ (|0\rangle + e^{2\pi i\cdot(2^{0}\omega)}|1\rangle)/\sqrt{2}\ )\ \ \ [\text{nth qubit}]$$

The factors written on the board are labelled 1st qubit, 2nd qubit, and nth qubit.

Why this works: write y in binary as $y = \Sigma_j\ y_j 2^j$. Then $e^{2\pi i\cdot\omega y} = \Pi_j\ e^{2\pi i\cdot(2^j\omega)y_j}$ and $|y\rangle = |y_{n-1}\rangle \otimes \ldots \otimes |y_0\rangle$, so the sum factors qubit by qubit. The qubit for bit y_j ends up in $(|0\rangle + e^{2\pi i\cdot(2^j\omega)}|1\rangle)/\sqrt{2}$.

Connection to Section 8: by the shifting rule, the phase on the 1st qubit is $e^{2\pi i\cdot(0.x_n)_2}$, which depends only on x_n. The 2nd qubit's phase depends on $x_{n-1}x_n$, and so on, until the nth qubit's phase involves all of $x_1 \ldots x_n$. Every qubit is an unentangled single-qubit state, which is what makes this state easy to work with, and for $n = 1$ it reduces to $H|x_1\rangle$ from Section 9.